TMG-GAN: Generative Adversarial Networks-Based Imbalanced Learning for Network Intrusion Detection

Hongwei Ding, Yu Sun, Nana Huang, Zhidong Shen, and Xiaohui Cui, *Member, IEEE*

***Abstract*— Internet of Things (IoT) devices are large in number, widely distributed, weak in protection ability, and vulnerable to various malicious attacks. Intrusion detection technology can provide good protection for network equipment. However, the normal traffic and abnormal traffic in the network are usually imbalanced. Imbalanced samples will seriously affect the performance of machine learning detection algorithm. Therefore, this paper proposes an intrusion detection method based on data augmentation, namely TMG-IDS. We name the proposed data augmentation model TMG-GAN, which is a data augmentation method based on generative adversarial networks (GAN). First, TMG-GAN has a multi-generator structure, which can be used to generate different types of attack data simultaneously. Second, we increase the classifier structure, which can optimize the generator and discriminator more efficiently based on the classification loss. Third, we calculate the cosine similarity between the generated samples and the original samples and other types of generated samples as a generator loss, which can further improve the quality of generated samples and reduce the class overlap area between the distributions of various generated samples. We conduct extensive experiments on two intrusion detection datasets, CICIDS2017 and UNSW-NB15. The experimental results show that compared with the advanced oversampling algorithm and the latest intrusion detection algorithm, the proposed TMG-IDS method has a good detection effect under the three indicators of Precision, Recall and F1-score.**

***Index Terms*— Internet of Things (IoT), intrusion detection, generative adversarial networks (GAN), TMG-GAN.**

I. Introduction

AT PRESENT, with the rapid development of big data, cloud computing and 5G communication technologies, the IoT has been widely used, making human life enter a new era of interconnection of all things. IoT connects the virtual network with the physical environment through advanced communication technology. While promoting the development of human intelligence, it also inevitably introduces the security risks and vulnerabilities existing in the virtual network into the physical environment. Therefore, how to ensure the security of IoT devices and reduce the threat of network attacks on enterprises and individuals has become a problem that the industry and network security technicians pay close attention to.

As an active defense technology that can effectively protect network security, intrusion detection has been widely studied. In recent years, many innovative intrusion detection technologies have been proposed [1], [2], [3]. Usually, researchers classify intrusion detection problems as binary classification or multi-classification problems. Based on this, many traditional machine learning classification methods have been used in the field of intrusion detection, such as decision tree, support vector machine, multilayer perceptron and random forest [4], [5]. However, traditional machine learning methods cannot effectively deal with massive, high-dimensional and complex network data due to their inherent characteristics. Deep learning methods can effectively learn the internal representation features of data, and have excellent effects on processing high-dimensional, complex and noisy data. Therefore, the application of deep learning techniques to the field of network intrusion detection has received extensive research. Deep learning techniques that are widely used in intrusion detection include models such as convolutional neural network (CNN) [6], recurrent neural network (RNN) [7], and autoencoder network (AE) [8].

However, although many advanced intrusion detection methods have been proposed by related scholars, there are still the following shortcomings. First of all, it is very difficult to collect effective attack samples from the network, and it takes a lot of manpower and material resources. This makes the number of attack samples far less than that of normal samples in the experiment, which leads to the problem of class imbalance in the data set. Secondly, in the face of a very small number of samples, it is difficult to achieve effective classification by relying on the traditional imbalanced data processing methods and deep learning methods.

Based on this, we propose a novel method to deal with imbalanced data for intrusion detection, which is called *Tabular Multi-Generator Generative Adversarial Network* (TMG-GAN). As shown in Fig. 1, we can observe the framework of the proposed TMG-IDS model and its potential application in intrusion detection. The upper part of Fig. 1 shows the training and prediction process of TMG-IDS model. The training process includes four three steps: (1) dataset

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Hongwei Ding, Nana Huang, Zhidong Shen, and Xiaohui Cui are with the School of Cyber Science and Engineering, Wuhan University, Wuhan 430000, China (e-mail: hwding@whu.edu.cn; nanahuang@whu.edu.cn; shenzd@whu.edu.cn; xcui@whu.edu.cn).

Yu Sun is with the School of Computing, National University of Singapore, Singapore 119077 (e-mail: sun.yu@u.nus.edu).

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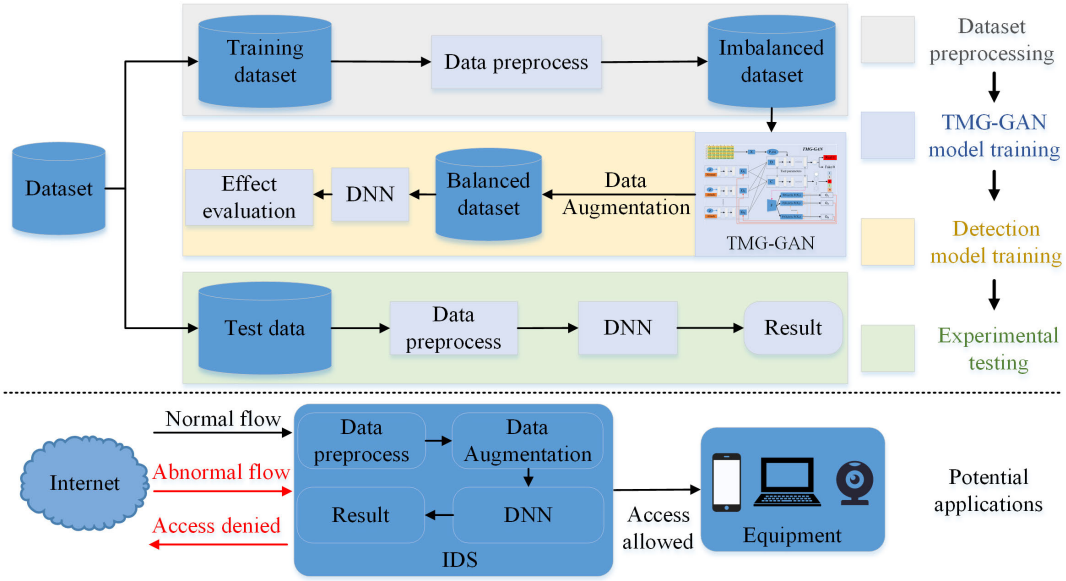


Fig. 1. TMG-IDS flowchart and application scenarios. The upper part shows the flow diagram of the proposed method, and the lower part gives an application example of the proposed research in network intrusion detection.

preprocessing; (2) training and data generation of the TMG-GAN model; (3) training the deep neural network (DNN) detection model using the balanced dataset. The test process is to use the DNN trained under the condition of balanced data set for final prediction. The lower part of the figure shows the potential application of our model in intrusion detection scenarios. Applying the deep learning model trained under the balanced data set to the real network environment can better resist network attacks. By setting a predetermined threshold in DNN, network traffic can be distinguished. Normal traffic can pass through the intrusion detection system, while abnormal traffic will be intercepted.

To sum up, our contributions are as follows:

- We propose a data enhancement scheme for imbalanced intrusion detection data, namely TMG-GAN model. In this model, we design a multi-generator structure, which can generate high-quality and low inter-class overlapping sample data at the same time.
- In order to improve the quality of the generated data, we add a classifier structure to TMG-GAN. The classifier is used to better distinguish various types of data, and the trained classifier can be used to filter noise data.
- In order to reduce the inter-class overlapping, we designed a feature extraction structure. By using the cosine similarity between high-dimensional features as the training loss of the generator, the distribution of various types of generated data can be better separated, thereby enhancing the clarity of the classification boundary.
- Experimental results on CICIDS2017 and UNSW-NB15 datasets show that our proposed TMG-IDS scheme can achieve more effective detection results than typical and state-of-the-art methods.
- As a data enhancement strategy, our method is not limited by the algorithm itself, so it can be combined with any advanced detection method, that is, TMG-GAN has strong universality.

The rest of the paper is organized as follows. In Section II, we review the related research work, including the intrusion detection methods under imbalanced data and the intrusion detection methods based on GAN. In Section III, the writing motivation is elaborated. In Section IV, the TMG-IDS method proposed in this paper is introduced in detail. In Section V, in order to evaluate the performance of TMG-IDS, sufficient experimental verification is made based on the existing data sets. In Section VI, the contents and experimental results of this study are summarized, and the future research work is prospected.

II. Related Work

In this section, we briefly summarize the network intrusion detection schemes for imbalanced data, mainly including the application of traditional imbalanced learning methods in network intrusion detection and imbalanced data processing methods based on GAN.

A. Application of Imbalanced Data Processing in IDS

Generally, the methods to solve the imbalanced learning problem in intrusion detection can be divided into three categories: data-level, algorithm-level and integrated learning methods.

The data-level method [9] aims at oversampling or undersampling the samples in the data set, so that the data set can reach a balance state before classification training. Oversampling [10] refers to generating more minority samples through data enhancement strategy, while undersampling [11] refers to reducing majority samples through data pruning strategy. The usual oversampling method [12] is to perform random linear interpolation between adjacent minority class samples

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to obtain new samples [13]. The undersampling operation removes samples that contribute less to the overall data distribution by quantifying the information of each majority class sample. For example, Tan et al. [14] proposed an intrusion detection method based on the combination of SMOTE and random forest to solve the problem of imbalanced learning in intrusion detection. Wang and Septian [15] proposed an intrusion detection framework combining oversampling technology and deep learning method. Firstly, the ensemble decision tree is used to select representative features; then, SMOTE oversampling technique is used to synthesize minority samples; finally, the RNN is used for classification and prediction. Al and Dener [16] proposed an intrusion detection method combining hybrid sampling and deep learning. In this method, a mixed sampling method combining SMOTE oversampling and Tomek-Links undersampling is used to balance the data set, and then a deep learning classifier combining CNN and long short-term memory network is used for final training and prediction.

The algorithm-level strategy [17] is to propose new methods or improve existing algorithms to adapt to the training of imbalanced data. The commonly used algorithm-level method is cost-sensitive learning. Cost-sensitive learning [18] is to integrate the misclassification cost of different types of data into the classification error, so as to better guide the classifier to modify the weight of minority samples in misclassification, and then increase the detection rate of minority samples. For example, Telikani and Gandomi [19] designed a cost-sensitive stacked auto-encoder (CSSAE) to deal with class imbalance in IDS. CSSAE allocates different cost to different types of data by generating cost matrix. Wang et al. [18] proposed an intrusion detection framework combining cost-sensitive learning and matrix learning. In this method, information entropy is introduced into matrix learning to measure the uncertainty of samples, so as to reduce the overall misclassification cost.

Ensemble learning method [20] uses multiple weak classifiers to make decisions together to enhance the final learning effect, so as to reduce the possible misclassification bias caused by a single classifier. Commonly used ensemble learning methods include data sampling-based ensemble and cost-sensitive ensemble. For example, Liu et al. [21] proposed an intrusion detection system combining ADASYN oversampling and LightGBM ensemble learning. ADASYN is used to synthesize attack samples of minority classes to solve the problem of data imbalance; LightGBM ensemble learning is used to improve detection accuracy and reduce system complexity. Gupta et al. [22] designed a three-tier IDS based on cost-sensitive and ensemble learning methods. The first layer uses cost-sensitive DNN to identify normal and abnormal flows; The second layer uses eXtreme Gradient Boosting to identify normal class, majority class attack and minority class attack; The third layer further classifies the minority attacks of the upper layer.

B. Generative Adversarial Networks

The generative adversarial network (as shown in Fig. 2) [23] is a deep generative model that fits the data distribution through the idea of adversarial learning, which includes two

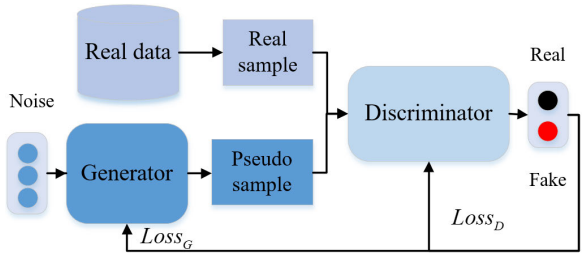


Fig. 2. Structure diagram of generative adversarial networks model.

structures of generator G and discriminator D. The generator is used to generate pseudo samples similar to the real distribution, and the discriminator is used to determine whether the input samples are real. The model is trained by alternating adversarial training, and the final generator can generate pseudo samples that approximately fit the real samples. Since GAN was proposed, it has been widely used in image generation, speech synthesis and text generation [24], [25]. Based on the good data distribution fitting ability of GAN, it has also been widely studied in data expansion and data enhancement [26]. In the field of network intrusion detection, some scholars also introduce GAN to solve the problem of data imbalance [27]. For example, Hao et al. [28] improved the existing GAN model by adding an encoder and modifying the loss function to make it more suitable for the generation of streaming discrete features, thereby solving the problem of insufficient minority class samples in network attack traffic. Ding et al. [29] proposed a TACGAN model suitable for tabular type data generation to address the data imbalance problem in intrusion detection. The generator in TACGAN adds two new losses, distance and similarity, which can generate high-quality new samples more efficiently. Andresini et al. [30] proposed a multi-stage intrusion detection framework that combines CNN and GAN, called MAGNETO. MAGNETO consists of three modules: image encoding, data augmentation and classification. The image encoding module converts the network data stream into visualized 2D images; the data augmentation module uses ACGAN to generate new samples; the classification module uses CNN to train and predict 2D images.

III. Motivation

Through the research on the solutions of imbalanced learning problems, data-level, algorithm-level and ensemble learning are faced with inherent problems. For the data-level method, the traditional oversampling method is difficult to effectively fit the overall distribution of data through local oversampling, and it is easy to introduce noise data. Undersampling method is not suitable for data with high imbalance rate, and it is easy to delete important samples, which makes classification more difficult. Algorithm level strategy needs to set specific misclassification cost matrix according to different application scenarios. The setting of this matrix is usually given by experts, so its applicability is poor. Ensemble learning needs to train multiple classifiers, and insufficient training samples for each classifier will lead to overfitting, which

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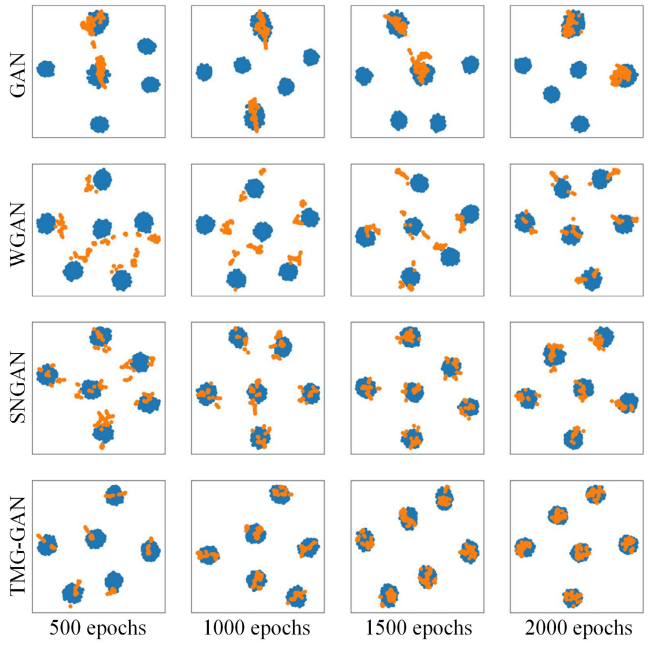


Fig. 3. Data fitting effects of different GAN models. The first is the original GAN model, the second and third are the improved classical models WGAN and SNGAN, and the fourth is the TMG-GAN proposed in this study. Blue data represents the distribution of the original samples, and orange data represents the distribution of the generated samples.

will affect the training effect. Therefore, ensemble learning methods are usually used in combination with data-level or algorithm-level strategies.

Although some scholars have used GAN in the field of intrusion detection, they still face the following difficulties. Firstly, the original GAN model is prone to model collapse, resulting in too single pseudo samples, so it is not suitable for the generation of multi-classification samples. Secondly, most of the current studies use the samples generated by GAN directly in the training of classifiers. Although this can expand the minority class samples, it may also introduce noise in the majority class samples. Thirdly, the imbalance between the majority class and the minority class is not the only factor that leads to the learning difficulties of the classifier. When there is inter-class overlapping in various data, the difficulty of classifier training will increase dramatically [31]. However, the current method does not consider whether the generated samples will increase the class overlap of samples. Based on this, we propose the TMG-GAN model to solve the above problems. We design a multi-generator and feature extraction structure for the problems of inter-class overlapping and mode collapse. Through the multi-generator structure and calculating the similarity loss between high-dimensional features of various types of data, not only can multiple types of pseudo samples be generated simultaneously, but also the sample distribution of various types of data can be effectively restricted. To solve the problem of noise samples, we designed a classifier structure. When the model training is completed, the classifier can be used as a noise filter, so that the generated data is easy to classify. Fig. 3 shows the data fitting effect of the four GAN models under adversarial learning. As can be seen from the figure, there are 6 data patterns in the original data. Under the same conditions, the TMG-GAN (using 6 generators) model can better capture the distribution of the 6 data patterns compared to the other three GAN models.

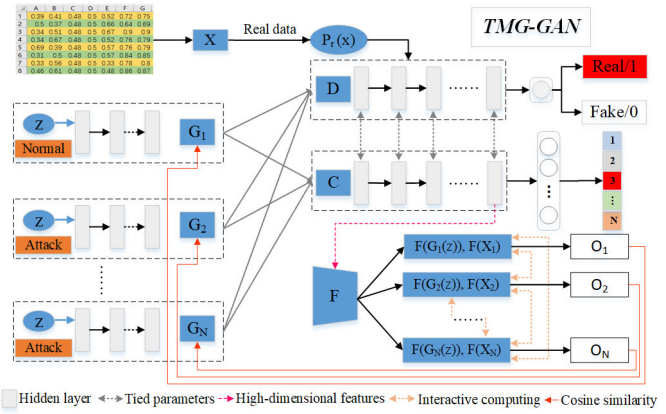


Fig. 4. Proposed TMG-GAN model structure diagram. $G_{1\sim N}$ represents the generator used to generate different types of data; D is the discriminator; C is the classifier; F is the feature extractor; $O_{1\sim N}$ is the calculated cosine similarity loss. $X_{1\sim N}$ represents samples of different classes in the original data. In addition, in this study, in order to reduce the complexity of the model, we tie the parameters of discriminator D and generator C, that is, the hidden layer of D and C shares the parameters.

IV. Proposed Methodology

Fig. 4 shows the structure of the proposed TMG-GAN model, which consists of four basic substructures: generator, discriminator, classifier, and feature extractor. The generator is used to generate pseudo sample data; the discriminator is used to judge whether the sample is real or fake; the classifier is used to judge the sample category; the feature extractor is used to extract high-dimensional features before the classification layer. We propose to use a multi-generator structure, which can process multiple types of data simultaneously, and combine feature extractors to generate samples with less class overlap. We also propose to add a classifier module, which can be used as a pseudo sample classifier and also as a noise filter. In addition, we also add a high-dimensional feature extractor to calculate the cosine similarity between each type of data and the original data and other types of data, so that the calculated cosine value is used to update the corresponding generator parameters.

A. TMG-GAN Model Training

This section introduces the detailed training process of TMG-GAN in detail, and the symbols used are shown in Table I. As shown in the table, where N represents the total number of data classes; X and $\tilde{X}$ represent the original sample set and the generated sample set respectively; G_k represents the k-th generator; D and C represent the discriminator and classifier respectively; m represents a The total number of samples in the batch; z represents Gaussian noise; x_k and $\tilde{x_k}$ represent the real samples and generated samples of the k-th class, respectively; $F(x)$ represents the high-dimensional features extracted by the sample x on the last hidden layer of the classifier; P_r and P_g represent the original Sample distribution and generating sample distribution; $O(F1, F2)$ represents the calculated cosine similarity value.

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TABLE I
DESCRIPTION OF SYMBOLS USED IN THIS SECTION

Symbol	Description
N	Total number of classes
X	Original sample set
$\tilde{X}$	Generate sample set
G_k	Generator for the k-th sample, $k \in \{1, 2, \cdots, N\}$
D	Discriminator
G	Generator
C	Classifier
m	Total number of samples in a batch
z	Gaussian noise
x_k	Real samples of class k
$\tilde{x_k} = G_k(z)$	Generated samples of the k-th class
$F(x)$	High-dimensional features extracted from sample x
P_r	Original data distribution
P_g	Generate data distribution
$O(F1, F2)$	Cosine similarity

1) GAN Model: GAN model is optimized through confrontation training, that is, G and D promote their opponents' continuous optimization through minimax game. The optimization purpose of G is to continuously promote the similarity between the generated distribution $p_g(z)$ and the original distribution $p_r(x)$. The optimization purpose of D is to accurately determine which distribution the input data belongs to. The following objective functions are used in GAN to train G and D simultaneously:

$$\min_G \max_D L(D, G) = E_{x\sim P_r}[\log(D(x))] + E_{\tilde{x}\sim P_g}[\log(1 - D(\tilde{x}))] \quad (1)$$

where x represents real data, $\tilde{x}$ represents pseudo sample data generated by generator G. where P_r is the true data distribution and P_g is the generative data distribution implicitly defined by $\tilde{x} = G(z)$, $z \sim P(z)$, for which z is sampled from a simple noise distribution P, such as a uniform, normal, or Gaussian distribution.

2) TMG-GAN Model: In this section, we mainly introduce the training of TMG-GAN model.

a) Cosine similarity calculation in TMG-GAN: As mentioned in the previous section, we regard the last hidden layer of the classifier C in TMG-GAN as extractor F of high-dimensional features. When C is fully trained, the high-dimensional features obtained by F can better distinguish the classes of the data, that is, the distribution of samples of different classes can be effectively identified. Based on this, we hope that the generated sample distribution of a certain class can be closer to the original sample distribution and different from the sample distribution of other classes of data. As shown in Fig. 5, we want the generated yellow samples to be closer to the original distribution ($Class_1$), and farther away from the distributions of the other classes of samples ($Class_{2\sim N}$). The specific implementation is as follows.

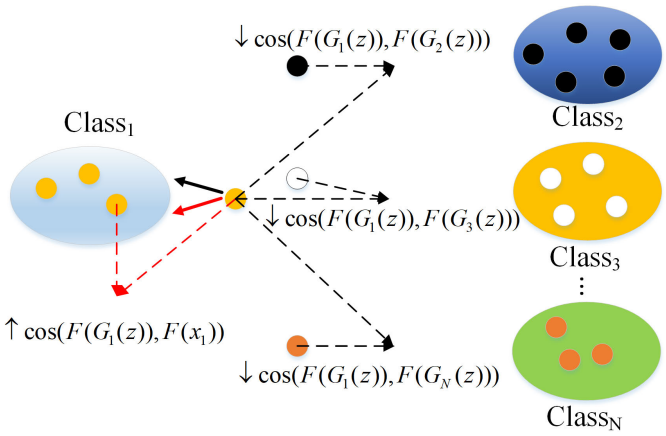


Fig. 5. Schematic diagram of cosine similarity applied in TMG-GAN. The red dashed line represents the cosine similarity between the original sample and the generated sample; the black dashed line represents the cosine similarity between the generated sample and the generated samples of other classes. The light blue area on the left represents the original distribution of the same type as the generated samples (yellow samples), and the area on the right represents the distribution of $N-1$ generated samples that are different from the yellow samples.

We assume that the sample generated by the generator is $\tilde{x_k} = G_k(z)$ and the corresponding original sample is x_k. After feature extraction by F, the high-dimensional features of the two samples are $F(G_k(z))$ and $F(x_k)$. Then the cosine similarity between the generated sample and the original sample can be obtained from the following equation:

$$\uparrow O_k(F(\tilde{x_k}), F(x_k)) = \left| \frac{F(\tilde{x_k})F(x_k)}{\|F(\tilde{x_k})\| \, \|F(x_k)\|} \right|, k \in \{1, \cdots, N\} \quad (2)$$

Similarly, the cosine similarity between the generated sample $\tilde{x_k} = G_k(z)$ and the generated sample $\tilde{x_j} = G_j(z)$, $j = \{1, 2, \cdots N\}$ *and* $j \neq k$ of other types can be obtained by the following equation:

$$\downarrow O_k(F(\tilde{x_k}), F(\tilde{x_j})) = \frac{1}{N-1} \sum_j \left| \frac{F(\tilde{x_k})F(\tilde{x_j})}{\|F(\tilde{x_k})\| \, \|F(\tilde{x_j})\|} \right|, \quad j \in \{1, \cdots, N\} \text{ and } j \neq k \quad (3)$$

Because we want the distribution of the generated sample $\tilde{x_k}$ to be closer to the original sample x_k, the greater the cosine similarity $O_k(F(\tilde{x_k}), F(x_k))$, the better. On the contrary, we hope that the distribution of generated sample $\tilde{x_k}$ will not coincide with the distribution of other types of generated sample $\tilde{x_j}$, so the smaller the cosine similarity $O_k(F(\tilde{x_k}), F(\tilde{x_j}))$, the better. Based on this, the cosine similarity loss finally used to update generator G_k is expressed as follows:

$$O_k = O_k(F(\tilde{x_k}), F(\tilde{x_j})) - O_k(F(\tilde{x_k}), F(x_k)), \quad \{k, j\} \in \{1, \cdots, N\} \text{ and } j \neq k \quad (4)$$

b) TMG-GAN model training: In TMG-GAN, we use a set of generators $G_{1\sim N}$ to jointly construct a mapping function that maps random noise z to data space X. The random noise z in TMG-GAN follows a Gaussian distribution $p_z(z)$, and the generator G_k samples from it to obtain a generative

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distribution $p_g(\tilde{x}_k)$. The distribution followed by the original data is $p_r(x)$. As an improved model of GAN, TMG-GAN still follows the minimax game of G and D, that is, generator $G_{1\sim N}$ is optimized towards minimization, while discriminator D is optimized towards maximization. In contrast to the original GAN model, each generator in TMG-GAN is encouraged to focus only on the generation of a certain class of samples and is not allowed to produce overlapping distributions with other classes of samples. Classifier C is used to identify the class of the input samples, and the classification loss is used to update the generation $G_{1\sim N}$ and the discriminator D. Based on this, the objective function of TMG-GAN can be expressed as follows:

$$\min_{G_{0\sim N}} \max_{D} = \frac{1}{N}\sum_{k=1}^{N}[E_{x_k\sim P_r}[D(x_k)] - E_{\tilde{x}_k\sim P_g}[D(\tilde{x}_k)] - E_{x_k\sim P_r}[C(x_k)] + E_{\tilde{x}_k\sim P_g}[C(\tilde{x}_k)]] + O_k \quad (5)$$

where $[E_{x\sim P_r}[D(x)] - E_{\tilde{x}\sim P_g}[D(G_j(z))]$ is the original objective function of WGAN [32]. $C(x_k)$ represents the classification loss used to update the discriminator D, and $C(\tilde{x}_k)$ represents the classification loss used to update the generator G_k. O_k represents the result of the cosine similarity calculated between the generated sample $\tilde{x}_k$ and other samples. The classifier C in TMG-GAN adopts the cross-entropy loss as the objective function, and its equation is as follows:

$$L_C = -\frac{1}{m}\sum_{i=1}^{m}\sum_{k=1}^{N}[y_{ik}\log(p_{ik})] \quad (6)$$

where y_{ik} and p_{ik} represent the true label and predicted probability that the i-th sample belongs to the k-th class, respectively. As can be seen from the above equations, the optimization formulas for D and $G_{1\sim N}$ are as follows:

$$\max_{D} = \frac{1}{N}\sum_{k=1}^{N}[E_{x_k\sim P_r}[D(x_k)] - E_{\tilde{x}_k\sim P_g}[D(\tilde{x}_k)] - E_{x_k\sim P_r}[C(x_k)]] \quad (7)$$

$$\min_{G_{0\sim N}} = \frac{1}{N}\sum_{k=1}^{N}[-E_{\tilde{x}_k\sim P_g}[D(\tilde{x}_k)] + E_{\tilde{x}_k\sim P_g}[C(\tilde{x}_k)]] + O_k \quad (8)$$

In addition, TMG-GAN draws on the idea of weight normalization in SNGAN [33]. By normalizing the weight matrix W, the gradient is controlled within the range of less than or equal to 1, thereby improving the original GAN model. The normalization of the weight matrix can be expressed as follows:

$$\bar{W} := {}^{W}/_{\sigma(W)} \quad (9)$$

By normalizing the W weight matrix to a formula whose spectral norm is always equal to 1, the gradient can be controlled to be less than or equal to 1, thereby further improving the problems of GAN training instability and mode collapse.

In Algorithm 1, we give the pseudocode for TMG-GAN training. The third line represents sampling various class training samples from real data, and the fifth line represents generating pseudo samples of various classes from generator G. Discriminator D is trained based on both real and pseudo samples to update the parameters, as shown in the sixth line. The tenth line indicates the re-generation of a batch of pseudo samples, combined with the designed cosine similarity loss (Eq. 4), for training generator G. The TMG-GAN model training is completed when the set maximum iteration t is reached.

Algorithm 1 Pseudocode for TMG-GAN Training

Input: Training sample set $X_{1\sim N}$; training times t_d of discriminator D and training times t_g of generator $G_{1\sim N}$ in each round of alternate training; round t of iterative training.

Output: Model parameters $(\theta^{(D)}, \theta^{(G)}, \theta^{(C)})$.

1: **while** item=t **do**
2: **for** $(p = 1; p \le t_d; p = p + 1)$ **do**
3: Sampling samples of different classes: $(x_k^i)_{i=1}^m \sim p_r(X_k)$
4: Sample $\{z^{(i)}\}_{i=1}^m \sim P_z$ a generated batch
5: Generate pseudo samples: $(\tilde{x}_k^i)_{i=1}^m = G_k(z^{(i)})$
6: $\nabla_{\theta(D,C)} = \frac{1}{m}\frac{1}{N}\sum_{i=1}^{m}\sum_{k=1}^{N}[D(x_k^i) - D(\tilde{x}_k^i) - C(x_k^i)]$
7: **end for**
8: **for** $(q = 1; q \le t_g; q = q + 1)$ **do**
9: Sample $\{z^{(i)}\}_{i=1}^m \sim P_z$ a generated batch
10: Generate pseudo samples: $(\tilde{x}_k^i)_{i=1}^m = G_k(z^{(i)})$
11: Calculate cosine similarity loss O_k using Eq. 4
12: $\nabla_{\theta(G_k)} = \frac{1}{m}\frac{1}{N}\sum_{i=1}^{m}\sum_{k=1}^{N}[-D(\tilde{x}_k^i) + C(\tilde{x}_k^i) + O_k^i]$
13: **end for**
14: item =item +1
15: **end while**
16: Return $(\theta^{(D)}, \theta^{(G)}, \theta^{(C)})$

B. Sample Generation

When the model training is completed, the generator $G_{1\sim N}$ in TMG-GAN can implicitly approximate the data distribution, that is, $G_{1\sim N}$ can be used to generate pseudo samples similar to the distribution of real positive samples. Here, we use a multi-generator to simultaneously generate multiple classes of attack sample data. Although our scheme is designed to generate high-quality sample data as much as possible, it may still produce a small number of noise samples. The noise data we defined in this study refers to those samples that may make the classifier make wrong judgments. Based on this, in order to further filter out higher quality data from the generated samples, we use the classifier as the filter of noise samples. When TMG-GAN is fully trained, C can better classify multi-class data, so C will be very sensitive to noise samples.

We assume that the various sample data generated by the generator $G_{1\sim N}$ is $\tilde{X}_{1\sim N}$, and the corresponding label is $Y_{1\sim N}$. We input the generated sample data $\tilde{x}_k$ into C, if $\tilde{x}_k \in \tilde{G}_k$ and $C(\tilde{x}_k) = \tilde{Y}_k$, $\tilde{x}_k$ belongs to the easy-to-classify samples; if $\tilde{x}_k \in \tilde{G}_k$ and $C(\tilde{x}_k) \neq \tilde{Y}_k$, then $\tilde{x}_k$ belongs to the difficult-to-classify samples. Based on this, we choose to retain the generated

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Algorithm 2 Attack Sample Generation Based on TMG-GAN

Input: The number $n_{1:k}$ of various attack samples to be generated.

Output: The generated attack sample set $\tilde{X}_{1\sim N}$.

// Call TMG-GAN trained in algorithm 1 to generate pseudo sample data
Sample $\{z^{(i)}\}_{i=1}^{m} \sim P_z$ a generated batch
Generate pseudo sample: $(\tilde{x}_k^i)_{i=1}^{m} = G_k(z^{(i)})$
// Noise sample filtering
$t_1 = 0, t_2 = 0, \ldots, t_N = 0$
while $t_k = n_k; k = 1, 2, \cdots, N$ **do**
    Using the trained classifier C to judge the label $C(\tilde{x}_k)$ of the pseudo sample $\tilde{x}_k$
    if $\tilde{x}_k \in \tilde{G}_kandC(\tilde{x}_k) == \tilde{Y}_k$ **then**
        Add $\tilde{x}_k$ to $\tilde{X}_k$
        $t_k = t_k + 1$
    end if
end while
Return $\tilde{X}_{1\sim N}$

samples that can be correctly classified by the classifier C in TMG-GAN, and filter out the difficult-to-classify pseudo samples. The specific data generation and filtering algorithm is shown in Algorithm 2.

V. Experiment

In this section, we will verify the effectiveness of TMG-IDS through experiments. Firstly, we give the detailed settings in the experiment process, including model parameters, evaluation criteria and data set introduction. Then, we do a series of experiments on the proposed intrusion detection framework. The comparative experiments mainly include: intrusion detection model based on classical oversampling method and advanced intrusion detection model. In addition, we also discuss the influence of imbalance rate on the experimental results. Finally, we designed ablation experiments to verify the effectiveness of all the improvements.

A. Experimental Setup

1) Data Sets: In the validation experiments, we use the standard datasets CICIDS2017 and UNSW-NB15 commonly used in intrusion detection. CICIDS2017 consists of 5 days of data collection, the first day only collected benign data, and the remaining four days included Brute Force FTP, Brute Force SSH, DoS, Heartbleed, Web Attack, Infiltration, Botnet and DDoS. Each instance in the dataset contains 78 feature attributes and 1 label attribute, including normal classes and common attack classes. The original UNSW-NB15 dataset contains 9 attack types, and each instance consists of 47 feature attributes and 1 label attribute. In this study, we choose 5 types of data for experiments, including normal data and 4 types of attack data. Among the four types of attack data, DoS and Reconnaissance represent common attacks in the dataset, and Shellcode and Worms represent rare attacks. The details of the training data and test data used in this paper are shown in Table II.

TABLE II
Dataset Information

Data sets	Label	Class	Training set	IR	OS
CICIDS2017	0	Benign	105222	-	105222
	1	DoS	21550	4.88	34609
	2	Port Scan	10809	9.73	23868
	3	Brute Force	5235	20.10	18294
	4	Web Attack	1476	71.19	14535
	5	Bot	857	121.78	13916
UNSW-NB15	0	Normal	56000	-	56000
	1	DoS	12264	4.57	22264
	2	Reconnaissance	10491	5.34	20491
	3	Shellcode	1133	49.43	11133
	4	Worms	130	430.77	10130

Note: "Label" represents the label assigned to each class of data; "Class" represents the class contained in the data set; "Training set" indicates the number of samples of each class in the training set; "IR" represents the ratio of normal samples to attack samples ; "OS" represents the number of instances in the training set after oversampling.

Due to the large difference in attribute values between the features of each dimension, we used the data normalization method to map the feature values of each dimension between 0.0 and 1.0. The data normalization formula is shown in Eq. 10.

$$x^* = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{10}$$

where x and x^* represent the original data and normalized data respectively; x_{min} and x_{max} represent the minimum and maximum values in the current attribute, respectively.

2) Implementation Detail: The detailed structure of the TMG-IDS[1] model is shown in Fig. 6. All generators in TMG-GAN keep the same settings, they all sample noise from the standard Gaussian distribution $N(0, 1)$. Five comparison methods are included in the baseline model, of which DNN is the most basic classification model. The purpose of setting the DNN is to observe the classification effect of the original data. Since ADASYN and SNGAN [33] are typical representatives of traditional oversampling and generative model oversampling, we use these two methods as baseline models. In addition, MAGENTO [30] and TACGAN-IDS [29] are recently published GAN-based intrusion detection methods, so we use these two methods as baseline models as well. For the generator and discriminator in SNGAN, we use the same structure as TMG-GAN; MAGENTO and TACGAN-IDS follow the structural setup described in the original paper. In addition, other conditions, such as training rounds (epochs = 2000) and experimental environment (Pytorch framework) are unified standards.

3) Assessment Metric: To more accurately evaluate the effectiveness of our method, we use three evaluation metrics, Precision, Recall, and F1-score. The results of the three evaluation indicators are calculated by the confusion matrix, as shown in Table III.

[1] https://github.com/universuen/TMG-GAN

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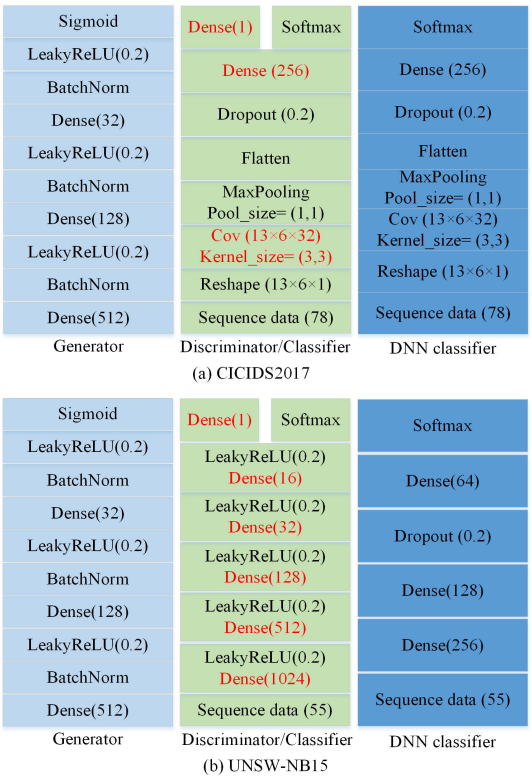


Fig. 6. Detailed structure of our proposed TMG-GAN model. Figures (a) and (b) are the model structures used to process CICDS2017 and UNSW-NB15 data sets respectively. The first column is the detailed structure of the generator; The second column contains discriminators and classifiers, in which the red mark indicates that the spectral normalization operation has been carried out; The third column is the model structure ultimately used to balance data classification.

TABLE III
CONFUSION MATRIX

Confusion Matrix		Predict	
		Attack	Normal
Real	Attack	True Positive (TP)	False Negative (FN)
	Normal	False Positive (FP)	True Negative (TN)

B. Experimental Result

In this section, we conduct extensive validation experiments using the CICIDS2017 and UNSW-NB15 datasets, including binary and multi-classification experiments.

1) Binary Classification Results: In the first validation experiment, we conducted a binary classification experiment on the preprocessed CICIDS2017 data set. In the experiment, we uniformly mark five kinds of abnormal traffic as attack samples. As shown in Table IV, we give the results of the normal data and attack data under the three evaluation indicators of Precision, Recall and F1-score, respectively. The "Macro-" in the last line is the macro average result of the three evaluation indicators, which is obtained by calculating the average of the results. The results in Table IV show that TMG-IDS has achieved the best results in the detection of normal and attack data. It can be seen from the macro average results that our method achieves 0.9973 on the three evaluation metrics of Precision, Recall and F1-score. Compared with traditional DNN, ADASYN, as a typical oversampling method, can improve the detection effect, but the improvement is limited. As a classical GAN model, SNGAN can also effectively improve the detection results. MAGENTO and TACGAN are two data expansion methods focusing on intrusion detection, so they can better improve the classification results. MAGENTO achieved the best results among the five baseline methods, and our method was further improved compared with this method (Precision, Recall and F1-score increased by 2.12%, 2.13% and 2.13%, respectively). In addition, compared with DNN, the results of TMG-IDS on the three evaluation indicators of Precision, Recall and F1-score are improved by 7.85%, 7.86% and 7.86%, respectively.

In Table V we present the experimental results for UNSW-NB15. The three deep generative models of SNGAN, MAGENTO and TACGAN can effectively improve the experimental results. Compared with the other five methods, TMG-IDS has achieved the optimal value, and compared with DNN's Precision, Recall and F1-score, it has increased by 8.18%, 3.46% and 7.20%, respectively. Based on this, it can be seen from the results of binary classification that our method achieves the optimal detection effect and can effectively detect normal and abnormal traffic.

2) Multi-Classification Results: As shown in Table VI, we give the multi-classification experimental results of CICIDS2017 data. It can be seen from the classification results that DNN has a certain detection effect on the three types of samples labeled "0", "1" and "2", while the three types of "3", "4" and "5" with a small sample size The detection effect of the samples is poor, especially the samples with the label "4" are extremely difficult to classify. Although ADASYN has a certain effect on the sample expansion of other classes, it is still difficult to achieve effective improvement for the fourth class of samples that are extremely difficult to classify (the Recall and F1-score values are only 0.1586 and 0.2725). SNGAN fits the sample distribution from a global perspective, so it is effective for the expansion of most samples. However, since there are only 857 samples in Category 5, it is difficult to obtain good training, so it cannot more effectively improve the detection performance of such samples (Recall and F1-score are 0.7696 and 0.8560, respectively). MAGENTO and TACGAN are two methods specially designed for intrusion detection. They have been improved in data preprocessing and model optimization respectively, which can effectively improve the quality of generated samples. Compared with other baseline methods, the detection effect of these two methods has been greatly improved. Table V shows that MAGENTO achieves the optimal values on Precision, Recall and F1-score, which are 0.9715, 0.9543 and 0.9623, respectively. Compared with the optimal MAGENTO model, TMG-IDS has been further improved by 2.48%, 4.38% and 3.49%, respectively.

In addition, the multi-classification experimental results of UNSW-NB15 are given in Table VII. The table shows that TMG-GAN has achieved the best values in the three evaluation indicators, which are 0.6677, 0.8238 and 0.7172, respectively. Compared with DNN, the results of TMG-GAN on Precision, Recall and F1-score increased by 6.21%, 16.38% and 15.92%, respectively. Based on this, it can be seen from the detection results of multi classification that TMG-IDS

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TABLE IV
BINARY CLASSIFICATION RESULTS (CICIDS2017)

Data	DNN			ADASYN			SNGAN		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
Normal	0.9139	0.9247	0.9193	0.9239	0.9485	0.9361	0.9266	0.9737	0.9496
Attack	0.9237	0.9126	0.9181	0.9470	0.9217	0.9342	0.9722	0.9227	0.9468
Macro-	0.9188	0.9187	0.9187	0.9355	0.9351	0.9351	0.9494	0.9482	0.9482
Data	MAGENTO			TACGAN-IDS			TMG-IDS (Ours)		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
Normal	0.9703	0.9821	0.9762	0.9940	0.9390	0.9657	0.9974	0.9973	0.9973
Attack	0.9818	0.9698	0.9758	0.9420	0.9943	0.9674	0.9973	0.9974	0.9973
Macro-	0.9761	0.9760	0.9760	0.9680	0.9666	0.9666	**0.9973**	**0.9973**	**0.9973**

Note: The best value of the result is shown in bold.

TABLE V
BINARY CLASSIFICATION RESULTS (UNSW-NB15)

Data	DNN			ADASYN			SNGAN		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
Normal	0.9935	0.8852	0.9362	0.9931	0.8768	0.9313	0.9960	0.9176	0.9552
Attack	0.6473	0.9731	0.7775	0.6306	0.9719	0.7649	0.7209	0.9831	0.8319
Macro-	0.8204	0.9292	0.8568	0.8119	0.9244	0.8481	0.8585	0.9504	0.8936
Data	MAGENTO			TACGAN-IDS			TMG-IDS (Ours)		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
Normal	0.9967	0.9381	0.9665	0.9957	0.9302	0.9619	0.9949	0.9503	0.9720
Attack	0.7750	0.9858	0.8677	0.7527	0.9816	0.8520	0.8096	0.9773	0.8856
Macro-	0.8858	0.9619	0.9170	0.8742	0.9559	0.9069	**0.9022**	**0.9638**	**0.9288**

Note: The best value of the result is shown in bold.

TABLE VI
MULTI-CLASSIFICATION RESULTS (CICIDS2017)

Data	DNN			ADASYN			SNGAN		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.8641	0.9744	0.9159	0.9248	0.9528	0.9386	0.9456	0.9902	0.9674
1	0.9386	0.9519	0.9452	0.9667	0.9417	0.9540	0.9959	0.9293	0.9615
2	0.9225	1.0000	0.9597	0.9561	0.9992	0.9772	1.0000	1.0000	1.0000
3	1.0000	0.4882	0.6561	0.8182	0.9956	0.8982	0.9831	0.9911	0.9871
4	0.7500	0.0085	0.0168	0.9655	0.1586	0.2725	0.9084	0.8286	0.8667
5	1.0000	0.4774	0.6463	0.9906	1.0000	0.9953	0.9643	0.7696	0.8560
Macro-	0.9125	0.6501	0.6900	0.9370	0.8413	0.8393	0.9662	0.9181	0.9398
Data	MAGENTO			TACGAN-IDS			TMG-IDS (Ours)		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.9672	0.9910	0.9790	0.9696	0.9766	0.9731	0.9994	0.9970	0.9982
1	0.9970	0.9607	0.9785	0.9724	0.9635	0.9679	0.9951	0.9983	0.9967
2	0.9977	1.0000	0.9988	0.9996	1.0000	0.9998	1.0000	0.9996	0.9998
3	0.9838	0.9911	0.9875	0.9745	0.9911	0.9828	1.0000	1.0000	1.0000
4	0.9328	0.8258	0.8760	0.9130	0.8626	0.8871	0.9833	0.9986	0.9909
5	0.9505	0.9572	0.9538	0.9462	0.9192	0.9325	1.0000	0.9952	0.9976
Macro-	0.9715	0.9543	0.9623	0.9626	0.9522	0.9572	**0.9963**	**0.9981**	**0.9972**

Note: The best value of the result is shown in bold.

obtains the optimal results compared with other methods, and can effectively detect normal traffic and various attack traffic.

C. Further Analysis

In this section, we further used the CICIDS2017 data set for experimental analysis, mainly including the experimental effect of our method under different sampling rates and the analysis of ablation experimental results.

1) Experimental Results With Different Sampling Ratios: In this part of the experiment, we will test the experimental results of TMG-IDS under different sampling ratios, and use this to verify the effectiveness and generalization of TMG-IDS in the face of a very small number of samples and extremely high imbalanced rates. The results of the previous multi-classification experiments show that the fourth type of data "Web Attack" is extremely difficult to classify compared to other types of data. Therefore, we take "Web Attack" as the experimental object and conduct binary classification experiments combined with normal types of data. Specifically, during the experiment, we train the TMG-GAN model by randomly sampling 10% 100% of the "Web Attack" data, combined with the complete normal data. The "Web attack"

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TABLE VII
MULTI-CLASSIFICATION RESULTS (UNSW-NB15)

Data	DNN			ADASYN			SNGAN		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.9915	0.9024	0.9449	0.9944	0.9095	0.9501	0.9930	0.9482	0.9700
1	0.7390	0.9124	0.8166	0.6374	0.8266	0.7198	0.7729	0.9230	0.8413
2	0.5253	0.8658	0.6539	0.5556	0.5315	0.5433	0.6916	0.6319	0.6604
3	0.2368	0.3201	0.2722	0.1328	0.8730	0.2305	0.1833	0.7619	0.2955
4	0.7778	0.1591	0.2642	0.4848	0.3636	0.4156	0.5714	0.3636	0.4444
Macro-	0.6541	0.6320	0.5904	0.5610	0.7008	0.5718	0.6424	0.7257	0.6423
Data	MAGENTO			TACGAN-IDS			TMG-IDS		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.9919	0.9492	0.9701	0.9928	0.9563	0.9742	0.9935	0.9498	0.9711
1	0.8103	0.9139	0.8590	0.8064	0.9298	0.8637	0.7944	0.9129	0.8496
2	0.7111	0.8581	0.7777	0.7623	0.6742	0.7155	0.6995	0.8616	0.7721
3	0.3877	0.7354	0.5078	0.2037	0.8122	0.3257	0.4527	0.7090	0.5526
4	0.3962	0.4773	0.4330	0.3333	0.4091	0.3673	0.6410	0.5682	0.6024
Macro-	0.6594	0.7868	0.7095	0.6197	0.7563	0.6493	**0.7162**	**0.8003**	**0.7496**

Note: The best value of the result is shown in bold.

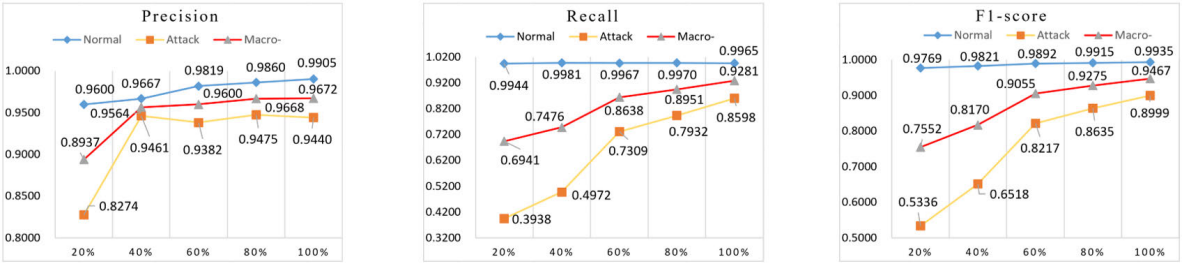


Fig. 7. Experimental results of raw data with different sampling ratios.

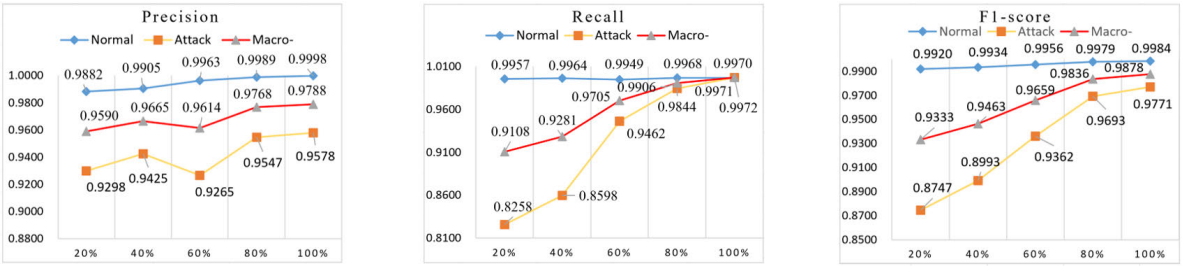


Fig. 8. Experimental results of TMG-IDS with different sampling ratios.

data under different sampling ratios are used for sample expansion, so as to achieve the balance between attack data and normal data. When 20% of the data is sampled from the complete sample, it can be calculated that the IR values of the normal sample and the attack sample reach an extremely high 356.38.

Fig. 7 and 8 show the results for the raw data, and the results for TMG-IDS, respectively. As shown in Fig. 8, it can be observed that TMG-IDS can achieve effective detection when using 100% of the Web Attack data (the macro-average results of Precision, Recall and F1-score are 0.9788, 0.9971 and 0.9878, respectively). When using 20% of the data, TMG-IDS can still achieve good detection results (the values of Precision, Recall and F1-score are 0.9590, 0.9108 and 0.9333, respectively). Despite the decreasing sampling ratio, the results of TMG-IDS for normal and attack classification are still relatively stable. Because TMG-GAN can obtain more attack samples with strong generalization ability by learning the distribution of minority class attack samples, thereby reducing the impact of the increase in the imbalance rate between normal samples and attack samples. In addition, as we expected, as the sampling ratio decreases, it can be observed that the detection performance of TMG-IDS gradually decreases. Because the data distribution of "Web Attack" becomes extremely incomplete when the sampling rate drops to a very low level, it is difficult for TMG-GAN to learn the complete distribution, which reduces the detection effect. However, it can also be observed from Fig. 8 that even though the sampling ratio is reduced to 20%, TMG-IDS still shows good performance for the detection of normal and attack samples. To sum up, by comparing the original data and the experimental results of TMG-IDS, our method shows high detection performance and good robustness despite the extremely high imbalance rate samples.

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TABLE VIII
ABLATION EXPERIMENT RESULTS

Combination	Multi-generator	Cosine loss	Filter	Evaluation indicators	0	1	2	3	4	5	Macro-
(1)	✗	✗	✗	Precision	0.9456	0.9959	1.0000	0.9831	0.9084	0.9643	0.9662
				Recall	0.9902	0.9293	1.0000	0.9911	0.8286	0.7696	0.9181
				F1-score	0.9674	0.9615	1.0000	0.9871	0.8667	0.8560	0.9398
(2)	✓	✗	✗	Precision	0.9480	0.9943	0.9950	0.9867	0.9283	0.9975	0.9750
				Recall	0.9911	0.9202	1.0000	0.9911	0.8258	0.9525	0.9468
				F1-score	0.9691	0.9558	0.9975	0.9889	0.8741	0.9745	0.9600
(3)	✓	✗	✓	Precision	0.9727	0.9856	1.0000	0.9824	0.9374	0.9879	0.9777
				Recall	0.9875	0.9730	0.9949	0.9911	0.8272	0.9715	0.9576
				F1-score	0.9800	0.9793	0.9975	0.9867	0.8789	0.9796	0.9670
(4)	✓	✓	✓	Precision	0.9994	0.9951	1.0000	1.0000	0.9833	1.0000	0.9963
				Recall	0.9970	0.9983	0.9996	1.0000	0.9986	0.9952	0.9981
				F1-score	0.9982	0.9967	0.9998	1.0000	0.9909	0.9976	0.9972

2) Ablation Experiment: We use ablation experiments in this section to verify the effectiveness of each improvement in the model. Table VIII shows the detection effects of different combinations of different improvements. Combination (1) indicates that there are no improved components, only the basic model SNGAN. After adding the multi-generator component in (2), the Precision, Recall, and F1-score under macro averages increase by about 0.88%, 2.87%, and 2.02%, respectively. Compared with (2), after adding the filtering operation in (3), Precision, Recall and F1-score increased by about 0.27%, 1.08% and 0.7%, respectively. (4) Compared with (3), which increases the cosine similarity loss, Precision, Recall, and F1-score increase by about 1.86%, 4.05%, and 3.02%, respectively. All improvements are included in (4), resulting in an optimal value. From the results of ablation experiments, the addition of each component in our method can effectively improve the performance of the final detection results. Based on this, we can see that every improvement in TMG-GAN is meaningful.

VI. CONCLUSION

Due to the continuous enrichment of the application scenarios of the IoT, the security risks and threats it faces are also increasing, and the security situation is becoming more and more severe. In this study, we propose an intrusion detection method for imbalanced data, called TMG-IDS, to improve the detection performance in the case of insufficient attack samples. Firstly, for the multi-classification problem in intrusion detection, a multi-generator structure is proposed, so that each generator can focus on the generation of a certain class of samples. The multi-generator structure helps to generate different classes of attack samples at the same time. Second, we add a classifier module, which can assist the generator to generate different classes of data on the one hand, and act as a noise sample filter on the other hand to reduce the generation of noisy samples. Then, based on the feature extractor, we also propose a cosine similarity loss for high-dimensional features of samples to reduce the inter-class overlap between generated samples. Lower inter-class overlap is more helpful in improving the clarity of classification boundaries. Finally, based on the balanced training samples, we design a DNN-based classifier that can more effectively improve the classification performance of the model. Experimental results on the generic and challenging CICIDS2017 and UNSW-NB15 datasets demonstrate that the proposed method outperforms state-of-the-art intrusion detection methods on both binary and multi-class classification. In addition, we further explore the generalization of the samples generated by TMG-GAN. The experimental results show that training the TMG-GAN model with only a few samples can also make TMG-IDS have a better detection effect on attack samples. Finally, we verify that the improvements in each part of the model are effective through ablation experiments. It is worth noting that what we provide is a data-level imbalanced processing method that is not limited by any detection algorithm, that is, the effective data generated by this method can provide support for any more advanced intrusion detection methods.

In the future, our next plan is to propose a generative adversarial network model based on neural architecture search (NAS). Using NAS to automatically design a group of network structures that can learn different patterns of data can better train the GAN model, resulting in higher-quality generated samples.

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Hongwei Ding is currently pursuing the Ph.D. degree with the Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, China, under the supervision of Prof. Xiaohui Cui. His research interests include data mining, image processing, generative adversarial networks, and imbalanced learning.



Yu Sun is currently pursuing the master's degree with the School of Computing, National University of Singapore, Singapore. His research interests include data mining, image processing, and generative adversarial networks.



Nana Huang is currently pursuing the Ph.D. degree with the Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, China, under the supervision of Prof. Ruimin Hu. Her research interests include data mining, user behavior pattern analysis, and recommender systems.



Zhidong Shen received the Ph.D. degree in computer software and theory from the School of Computer Science, Wuhan University, Wuhan, China, in 2006. He is currently an Associate Professor with the School of Cyber Science and Engineering, Wuhan University. His research interests include artificial intelligence security, system security, and big data analysis.



Xiaohui Cui (Member, IEEE) is a Professor with the School of Cyber Science and Engineering, Wuhan University, China. He has published more than a hundred research papers in international journals and conferences (e.g., IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE TRANSACTIONS ON CYBERNETICS, TR, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *IPM*, PR, and ACL). He has published extensively and regularly contributes to leading academic journals and conferences in his field. His main research interests include big data, natural language processing, blockchain technology, food safety, and image processing.

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